

Newborn Health Risk Assessment

Input and Results Summary

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Important: This educational report is not a medical diagnosis and does not replace professional care.

Plain-Language Result

Risk Level MODERATE

A few factors in the information entered may need closer attention and follow-up.

Main Things to Notice

Baby was born earlier than expected. - Birth weight is lower than the usual range.

What You Should Do

Arrange or continue follow-up with a healthcare professional. - Do not rely on this result alone for medical decisions.

When to Seek Urgent Medical Help

The baby has serious difficulty breathing. - The baby is difficult to wake or is not feeding normally.

General warning signs follow World Health Organization and NHS newborn-care guidance.

Assessment Details

Input Summary

Gestational age	36.5 weeks
Birth weight	2.35 kg (2350g)
Mother's age	31 years
Child's gender	Female
Delivery method	Cesarean Section
Delivery complication	Yes
Known family disease	Yes
Disease name	Congenital Deafness
Who has it	Other family member

APGAR Summary

Component	Score	Interpretation
Appearance	1 / 2	Pink body, blue hands/feet
Pulse	2 / 2	Strong (100+ bpm)
Grimace	1 / 2	Grimace only
Activity	2 / 2	Active movement
Respiration	1 / 2	Weak or irregular
Calculated total	7 / 10	Normal (7-10)

Fuzzy Risk Module Results

Risk module	Index	Level
Immediate Condition Risk	41.2 / 100	Moderate
Birth-Related Risk	63.8 / 100	Moderate
Family-History Risk	45.0 / 100	Moderate

Family-History Details

Condition	Who has the disease	Risk Index
Congenital Deafness	Other family member	45.0 / 100

Assessment Factors

Main Contributing Factors	Why it appears
1. Preterm gestational age	Gestational-age membership is strongest in the preterm set.
2. Low birth weight	Birth-weight membership is strongest in the low set.

Lower-Impact Factors

Typical maternal age range

Confidence Basis

All important inputs were complete and valid. The final fuzzy levels showed a usable but not strong separation.

Important Triggered Fuzzy Rules

Activation is a rule firing strength from 0 to 1, not a contribution percentage.

Rule ID	Rule name	Description	Result	Activation
BR-06	Preterm with low birth weight	Preterm gestation together with low birth weight elevates birth-related risk.	High	0.75 / 1.00

Overall Hierarchical Fuzzy Result

65.4 / 100	Risk Level: Moderate Confidence: Moderate Arrange or continue follow-up with a healthcare professional and pay closer attention to the highlighted factors.
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The final index is produced by hierarchical fuzzy rules using Immediate Condition Risk, Birth-Related Risk, and Family-History Risk. It is not a fixed weighted average.